

Classifying Sexual Education Material with Generative AI*

Understanding how generative models classify the age appropriateness of sentence excerpts from sexual education material

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While sexual education is a critical part of child development, there is debate around how and when it should be taught. Social media companies use large language models tools to flag content that is inappropriate for younger audiences. Using zero-shot learning, this report examines how generative AI classifies the age appropriateness of sexual education material in order to understand what language factors into this decision. Model performances are then evaluated based on how they compare to the general model consensus.

1 Introduction

Due to the subject matter, this paper contains sensitive language that may be considered offensive.

The state of sexuality education in schools varies around the world. The topic of if, when, and how to introduce children to certain subjects is divisive, and often raises religious, political, and moral issues. According to the World Health Organization (2026), comprehensive sexuality education is defined as “a curriculum-based, incremental and scientifically-accurate approach that teaches children and young people about the cognitive, emotional, physical and social aspects of sexuality in a culturally-relevant and age-appropriate way.” That being said, what is considered age-appropriate for a given demographic is hard to define. Even if it could be universally agreed upon that a piece of content was inappropriate for children to consume, the exact reasons why may be hard to put into words. The exact criteria often comes down to what United States Supreme Court Justice Potter Stewart once said when asked to define obscene pornography: “I know it when I see it” (Lattman 2007).

*Code and data are available at: https://github.com/rosiet1039/sex_ed_classification_paper

According to a 2024 report, 66,000 Instagram posts, 347,200 Tweets, and 1.7 million Facebook posts were shared every minute of the year (Domo 2024). Many of these pieces of content may contain sensitive themes. Websites such as Youtube or TikTok are frequented by children seeking educational information (Döring 2024). In the early 2000s, before censorship laws were put in place, the internet was essentially unregulated, allowing anyone with internet access to consume and post offensive, violent, and pornographic material. Over the years, various countries have put legislation in place to help hide sensitive content from younger users, such as Canada’s recent Bill C-34, which mandates the implementation of safeguards on social media sites and AI chatbots in order to protect children from harmful material (Miller 2026). The detection of inappropriate content on websites frequented by minors has thus become a legal necessity, making research in the area of data classification of great interest to social media companies. Seeing as it is virtually impossible for humans to manually check content at this volume, Justice Stewart’s criteria falls short in today’s world.

Instead, companies perform this detection using a mix of algorithms and user reports. In 2021, Meta announced they would begin using artificial intelligence models trained with few-shot learning to detect harmful content on their platforms (Meta 2021). Few-shot learning is a technique in which a model is provided with some small number of demonstrative examples before a task is given (Brown et al. 2020). This is a common method used to help improve the accuracy of LLM text classification. Text classification is an important area of natural language processing, a subfield of computer science that focuses on getting computers to work with human language. Traditional methods for representing text ignored semantic meaning, and would, for example, treat the word “tie” the same whether it referred to the accessory or an equal score (Altinel and Ganiz 2018). Newer models have overcome this limitation and are able to not only discern between homonyms, but understand tonal differences in text that can be easily picked up on by humans but are very difficult to define. This has vastly broadened the use cases for artificial intelligence.

With the recent advances in semantic text classification technology, much research is being done attempting to use natural language processors as models for human behavior and attitudes. The training corpus for many leading LLMs is essentially the whole internet, containing every human thought ever published online. This creates an interesting opportunity to see how well model responses reflect human populations.

It’s Perfectly Normal is a ten-and-up sexual education book by Robie Harris and Michael Emberley (Harris and Emberley 2014). Though it has been praised for its detail and comprehensiveness, the book has gained criticism from those who argue that its contents are not appropriate for the target demographic (Perring 2005) (Hersher 2014). A random sample of 100 sentences was taken from the book, and using zero-shot learning, five different models were prompted to classify whether or not each text excerpt is appropriate for ages ten and up.

Model responses were then examined using scatterplots of two dimensional embeddings of each sentence. Sentences that discussed topics such as arousal, puberty, external genitalia, and sexual health tended to be flagged as inappropriate most often by all models. Performances were evaluated using alternative recall and precision metrics which compare individual classifications

to overall consensus. It was found that GPT 5.4 Mini had the highest recall (79.0%) and the lowest precision (35.4%), while Claude Haiku had the inverse, (with scores 21.3% and 51.3% and respectively). GPT 4o had the best F score (49.3).

This rest of this report will be separated into the following sections: Section 2 will provide background information on the key areas most relevant to this paper. Section 3 will go through relevant literature and related analyses. Section 4 will showcase the data used in the analysis. Section 5 will describe the models and prompting technique. Section 6 will detail the results and findings of the analysis. Section 7 will discuss the big picture relevance of the results.

2 Background

The content of this paper involves three main subjects: (1) childhood sexuality education, (2) online censorship, and (3) semantic text classification. Much research has been done in all three areas that have greatly informed and inspired this report.

2.1 Childhood Sexuality Education

Legislation regarding sex education varies greatly around the world, even within countries such as Canada and the United States. An extreme example of this is the United States, which is known for its great political divide. SEICUS, an organization dedicated to promoting healthy sexual education in the US, has created detailed state profiles that go in depth on each region's position on how sex ed should be delivered to students (SEICUS 2025). As of 2025, 14 states provide abstinence-only education, which has been shown to fail in protecting students from sexually transmitted illness and pregnancy (Santelli et al. 2006). 19 states exclude and/or stigmatize LGBTQ+ people in their sex education. Only three states, California; Oregon and Washington, require comprehensive sexuality education in all schools.

Different areas have vastly different ideas of what is age-appropriate for those under the age of 18 to learn about. Parents and guardians may have a desire to protect their children from certain subjects for fear that learning about sexuality and gender will increase risky behaviour, however there is much evidence to the contrary (World Health Organization 2026). A paper on parental attitudes towards the K-12 gender and sexuality curriculum in Australia identified that many believe children are too immature and easily influenced to learn about these topics (Ferfolja, Manlik, and Ullman 2024). Regardless of age-appropriateness, young people may be exposed to sexual concepts through the internet, peers, and real life experiences. Without a reliable foundation of knowledge from which to draw from, they may be unable to tell myth from truth.

2.2 Online Censorship and Age Restriction

As the internet and social media gain increasing popularity, countries have each put in place legislation to regulate what kind of content can be posted online. The Online Streaming Act (Bill C-11) of 2022 gave the Canadian Radio and Television Commission the ability to regulate Canadian digital media platforms the same way they had been regulating radio and TV broadcasters (Woolf 2022). This bill received pushback from those who hold the opinion that online censorship infringes on human rights (Lewis 2023). Regardless, many social media platforms are getting increasingly strict with their terms of service due to financial and legal pressure, such as content platform OnlyFans banning sexually explicit content at the request of credit card companies (cnn 2021) (Liao 2018). The Canadian government has stated that they intend to implement a social media ban for those under the age of 16, with an exemption for platforms that have sufficient safeguards in place (Miller 2026). These safeguards must protect children from viewing “harmful content”, which is defined as media that falls into any of these categories:

- Intimate content communicated without consent
- Content that sexually victimizes a child or revictimizes a survivor
- Content that induces a child to harm themselves
- Content used to bully a child
- Content that foments hatred
- Content that incites violence
- Terrorism or violent extremism content

Unfortunately many of these types of content cannot be fully accurately detected by a simple algorithm. There is no one word that is in every piece of hate speech, no single phrase that consistently promotes violence. Platforms often use user reports to aid algorithms in this detection, however this opens the door to unwarranted mass reporting performed by bots or spiteful users (Contreras 2021). Content by LGBTQ+ individuals is often disproportionately suppressed, and sexual education is usually detected as explicit (Tremblay 2026) (York 2024) (Zhang and Zou 2025). Additionally, many posts that arguably do fit the criteria go unremoved (Seitz 2022). It would be impossible for a healthy sample of humans to go through every social media post and assess potential harm. It is in the best interest of companies to build more intelligent algorithms that are able to detect nuanced semantics in order to keep online spaces safe and inclusive.

2.3 Text Classification

Text classification is a common task in machine learning in which a model has to assign a category or label to a piece of text. This category could be the topic the text is relating to, the sentiment, the tone, or something else entirely. There are many different techniques used to complete this task.

The majority of text classification performed today is supervised (Stryker, n.d.). In this method, sets of pre-classified text (which have been labeled by humans) are fed to the model in order to train it. The model uses these examples to get an idea of what goes in each category. It is then given new pieces of text which it must classify according to this same pattern.

Text clustering is a type of unsupervised text classification, a method where instead of training the model on labeled examples, it finds patterns within the data. This technique essentially puts texts into groups based on semantic similarity. Texts within the same cluster are assumed to relate to the same topic. After this is performed, the clusters can be inspected to determine which topic they each refer to.

Zero-shot, one-shot, and few-shot learning are all prompting techniques used to improve the ability of a pre-trained model to perform a specified task. The model is first given some number of completed examples of said task, and then asked to perform it on new data. Few-shot refers to multiple examples, one-shot refers to one, and zero-shot asks the model to execute the task based solely on what it already knows. In the context of text classification, this would look like simply prompting the model to classify some piece of text, usually into a specified set of categories. Zero-shot is often used when there is no labeled data available with which to provide examples. While this method has less accuracy than one- and few-shot learning, it still proves to be quite successful, with GPT-3 achieving 64.3% accuracy on TriviaQA, a dataset of question and answer problems, and 76% on LAMBADA, a dataset used to test word prediction capabilities (Brown et al. 2020).

3 Literature Review

4 Data

4.1 Overview

The data analyzed in this paper comes from the 2014 edition of the book *It's Perfectly Normal* by Robie Harris and Michael Emberley. All handling of data was done using the statistical programming language R (R Core Team 2025) as well as Python version 3.13 (Python Software Foundation 2026a). R packages `readr` (Wickham, Hester, and Bryan 2024), `text` (Kjell, Giorgi, and Schwartz 2023), `tidyverse` (Wickham et al. 2019), and `uwot` (Melville 2025), and Python packages `pymupdf` (McKie and Giese 2026), `nltk` (Bird and Loper 2004), `re` (Python Software Foundation 2026b), and `pandas` (team 2020) were used to clean and analyze data. R package `ellmer` (Wickham et al. 2026) was used to access the models used in this analysis. All plots were created using `ggplot2` (Wickham 2016), `ggrepel` (Slowikowski 2024), and `ggpubr` (Kassambara 2026). `knitr` (Xie 2014) and `kableExtra` (Zhu 2024) were used to help format this document.

4.2 *It’s Perfectly Normal*

The data used in this analysis is a random sample of sentences collected from the most recent edition of the book *It’s Perfectly Normal* by Robie Harris and Michael Emberley, published in 2014 (Harris and Emberley 2014). The first edition of this book was originally published in 1994 and has since been updated three times (Flynn 2019). *It’s Perfectly Normal* is a sexual education book intended for children ages 10 and up, and has been praised for its clear language, comprehensive information, and discussion of the LGBTQ+ community (Perring 2005). Despite this, the picture book has also gained widespread controversy for its detailed illustrations and themes which some consider to be inappropriate for the target audience (Hersher 2014). It has frequently been in the top 10 of the American Library Association’s list of most challenged books since its original release over two decades ago (American Library Association 2007).

4.3 Cleaning

In order to use *It’s Perfectly Normal*, all text was extracted from a pdf of the book and parsed into sentences, which were then written into a csv file. Only main body text was included. While much effort was taken to preserve the book’s intended text format, there were a small number of incomplete sentences in the sample. These were manually completed by referencing

5 Methods

Several models were used to examine how generative LLMs classify text-based sexual education material. Default parameters were maintained on all models except for temperature. The temperature of a model indicates the variability of its responses, with 0 being the least variable and 1 the most. This parameter was set to 0.4 in order to examine how model response varies. The models used were Open AI’s GPT 4o mini and GPT 5.4 mini, Deepseek’s V4 Flash and V4 Pro, and Claude Haiku 4.5 (OpenAI et al. 2024) (Singh et al. 2026) (DeepSeek-AI 2026) (anthropic 2025).

In order to prompt the models, zero-shot learning was employed. Zero-shot refers to a prompting style where no demonstrative examples are provided, and the model must base its answer on previous knowledge. 100 unique sentence excerpts from *It’s Perfectly Normal* were provided to each model, in the following format: ‘Is the following text in quotes appropriate for children ages 10 and up? Answer yes or no [sentence in quotations]’. Models then provide a yes or no response. Each text excerpt was provided to each model 30 times, and responses were recorded.

6 Results

6.1 Semantic Embedding and Clustering

In order to understand what kind of language tended to be classified as inappropriate, semantic text embeddings were created for each sentence. This refers to the process of translating text into a long list of numbers called a vector, which points to an area in multi-dimensional space. This translation takes into account semantic meaning, rather than just the words themselves, so sentences that mean similar things will be located closer together. This was done using function ‘textEmbed’ from the ‘text’ package in R, with model bert-case-uncased, which can be found for free on HuggingFace (Kjell, Giorgi, and Schwartz 2023) (Devlin et al. 2018).

The sentences were then divided into clusters using the Hartigan and Wong (1979) K-means algorithm. This algorithm attempts to find the optimal way to divide data into some pre specified k groups called clusters. This grouping is done based on the Euclidean distance (straight length) between points. In this case, clustering helps to identify the more specific themes at play within the sample of sentences. K=3 was chosen for the number of clusters. Through manual inspection, the following themes were identified:

1: Myths, Life Advice, and Internet Safety 2: Healthcare, STIs, Arousal, Puberty, and External Anatomy 3: Internal Biology and Anatomy

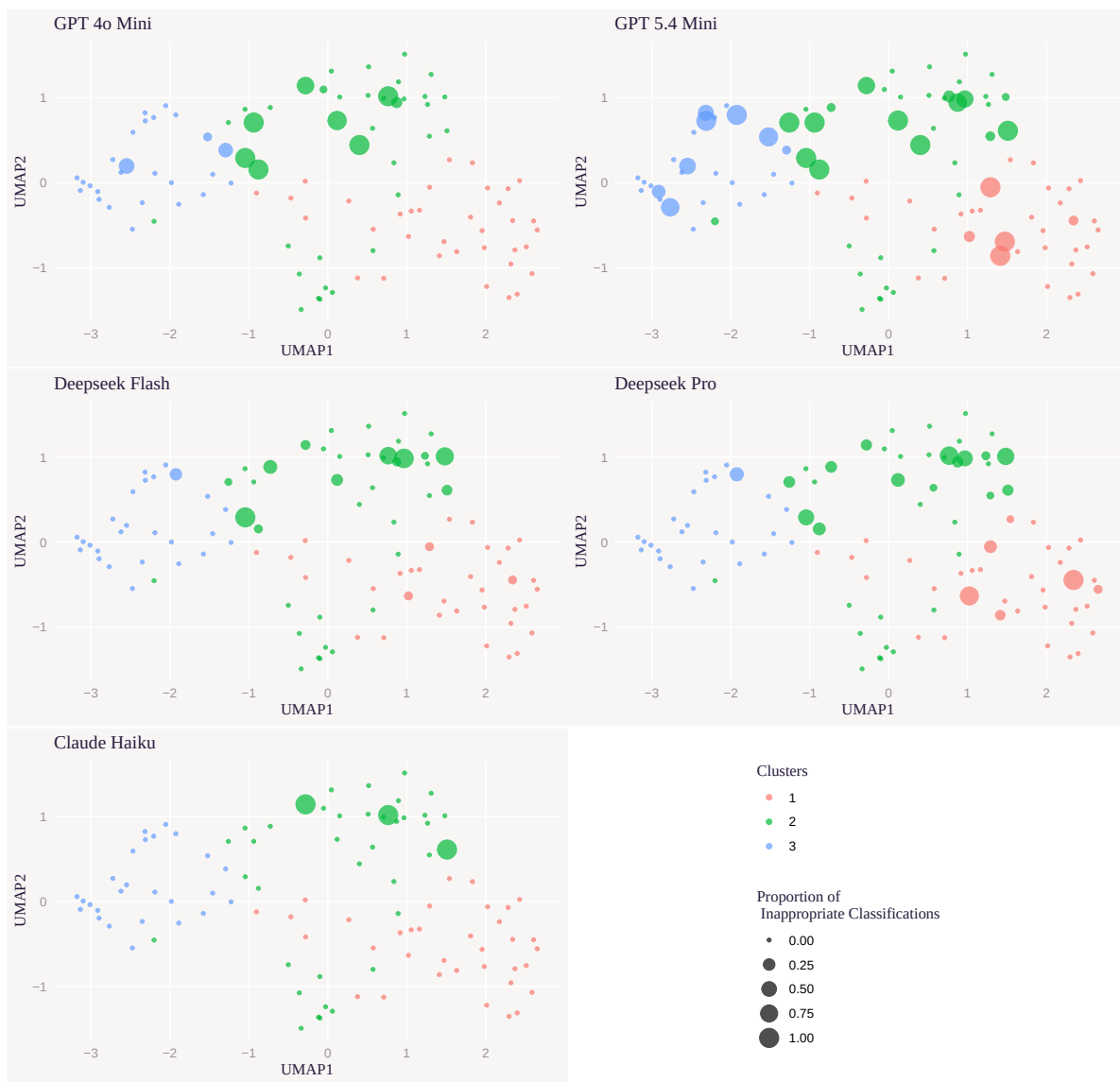
[Question for Rohan: Is there a more objective way to chose the number of clusters? I picked 3 because it produced the clearest themes/patterns in each cluster. Also- is there a more reliable way to identify said themes? I simply identified them by reading the sentences in each cluster. Would it be more “objective” if I fed some sample of sentences in each cluster to a pretrained model and asked it to identify themes?]

The embeddings were then compressed into two dimensions using Uniform Manifold Approximation and Projection (UMAP), a dimension reduction technique, with cosine similarity (angular distance between vectors) as the chosen distance metric. This was done in order to visualize the embeddings in the scatterplots below:

6.2 Performance Metrics

Due to a lack of reliable labelled data, model responses could not be compared to a ground truth. Model performances were thus measured using alternative precision and recall metrics found in Fedorchuk and Lamiroy (2017), which compare individual classifications to overall model consensus. Consider the probability $P(\delta_i)$ of sentence δ_i to be classified as inappropriate (model response “No”). Then:

$$P(\delta_i) = \sum_{k=1}^s \frac{S_k(\delta_i)}{s}$$



Where

$$S_k(\delta_i) = \begin{cases} 0 & \text{if } Yes \\ 1 & \text{if } No \end{cases}$$

is the classification result from model S_k of sentence δ_i , and s is the total number of models

Assuming sentences are evenly distributed, valid probabilistic measures of precision $Pr(S_k)$ and recall $Rc(S_k)$ are as follows:

$$Pr(S_k) = \sum_{i=1}^d \frac{S_k(\delta_i)P(\delta_i)}{S_k(\delta_i)} Rc(S_k) = \sum_{i=1}^d \frac{S_k(\delta_i)P(\delta_i)}{P(\delta_i)}$$

Then the F score is the harmonic mean of these two metrics:

$$F(S_k) = \frac{2Pr(S_k)Rc(S_k)}{Pr(S_k) + Rc(S_k)}$$

These statistics were calculated for each model 30 times across the 30 trials. The (arithmetic) mean and standard deviation of each metric is shown below.

Model	Metric	Mean (%)	Standard Deviation (%)
GPT 4o Mini	Recall	48.4	5.5
	Precision	50.6	4.5
	F Score	49.3	4.0
GPT 5.4 Mini	Recall	79.0	6.1
	Precision	35.4	2.3
	F Score	48.9	2.8
Deepseek V4 Flash	Recall	30.9	8.5
	Precision	53.9	6.4
	F Score	38.6	8.3
Deepseek V4 Pro	Recall	38.3	10.9
	Precision	45.2	6.8
	F Score	40.8	8.2
Claude Haiku 4.5	Recall	21.3	2.6
	Precision	57.3	7.9
	F Score	31.1	3.7

7 Discussion

As the sentence embedding scatterplots demonstrate, models all tended to classify topics as inappropriate in the second cluster (highlighted in green) more than the other two. Manual inspection of the texts in this cluster showed they pertain to sexual health, arousal, puberty, and external anatomy. A few examples of sentences in each cluster are below, as well as statistics showing how much of each cluster was classified as inappropriate by each model.

[insert examples and table of statistics]

[insert discussion of how this relates to human attitudes on sexual education, current curriculum around the world]

Due to a lack of ground truth data, model performances were measured using recall and precision metrics that compare responses to the overall model consensus. According to these statistics, GPT 5.4 had the highest mean recall (79.0%) and lowest mean precision (35.4%), meaning it correctly classified the most inappropriate sentences, but incorrectly classified the most appropriate ones. Claude Haiku demonstrated the opposite behaviour, with the lowest mean recall (21.3%) and highest mean precision (57.3%). This model was much more conservative when classifying text as inappropriate, only putting three sentences in this category out of 100 prompts and 30 trials for each prompt. It should be noted that Claude Haiku had no variance in its responses, and always responded with Yes or No for each unique prompt. GPT 4o Mini had the highest mean F score (49.3%), demonstrating the most balanced behaviour overall, while Claude Haiku had the lowest (31.1%).

All of these scores fell within the range of 30% to 60%, with the exception of GPT 5.4 Mini's recall score of 79.0%. Rather than indicating poor performance across all models, as they would if these metrics were based on ground truth, they demonstrate a lack of model consensus on the proper threshold for what content is age appropriate for children ten and older. [comment on how this aligns with varying human attitudes around sex ed]

[add conclusion]

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